ANKUSH MANHAS

AI Trainer and Data Annotator | AI Model Evaluation

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PROFILE

AI Trainer and Data Annotator who evaluates AI-generated responses for accuracy, relevance, instruction following, and clarity against hand-verified ground-truth keys. Built a 24-item benchmark for a financial RAG agent with source citations, and audited a 24.4M-row training dataset, catching label leakage and concept drift (recovered top-1,000 fraud detection from 17 to 150). Computer Science graduate (2026); fluent in English, Hindi, and Punjabi.

TECHNICAL SKILLS

- **AI Evaluation and Data Quality:** Response Quality Assessment, Ground-Truth Benchmark Design, Human-in-the-Loop Review, Model Evaluation, Data Annotation, Data Labeling, Factual Accuracy Verification, Instruction Following, Error Analysis, Quality Assurance, Data Validation, Annotation Guidelines
- **AI Assistants:** ChatGPT, Google Gemini, Claude, Perplexity, Prompt Engineering, Multi-Turn Conversations, Structured/JSON Output
- **Programming:** Python (pandas, NumPy, scikit-learn), SQL (SQLite), FastAPI, Streamlit, Docker, Git/GitHub, GitHub Actions CI
- **Machine Learning and NLP:** RAG, LangChain, LangGraph, embeddings (sentence-transformers: all-MiniLM, bge, all-mpnet), BM25 + reranking, XGBoost, logistic regression, text classification, drift analysis

PROJECTS

FinSight — AI Response Evaluation Benchmark for a Financial RAG Agent

Feb 2026 to Present | Python, LangChain, LangGraph, Gemini, sentence-transformers

- **Accuracy evaluation:** Evaluate AI answers against a hand-verified key of 24 financial metrics (revenue, margin, D/E, ROE, EPS), each tagged to its source PDF page; no answer accepted without a citation.
- **Instruction following & A/B output comparison:** Compared 4 embedding models and query variants; query expansion ("PAT" → "profit after tax") scored 0.55 vs 0.08, beating a larger model (0.35).
- **Annotation guidelines:** Wrote guidelines for PDF failure modes (₹ mis-decoding, million vs crore, Indian digit grouping) so label errors are caught early.

Training-Data Audit — Real-Time Fraud Detection (24.4M transactions, deployed)

2025 to 2026 | Python, scikit-learn, Kafka, Redis, FastAPI, Docker, Evidently

- **Label error & leakage detection:** Found label leakage in PaySim (receiver balances never updated for fraud: 49.8% vs 0.06%; 6.35M of 6.36M senders never repeat) and led the switch to IBM TabFormer (24.4M rows).
- **Error analysis at scale:** On 180,000 live-scored unseen purchases, caught 33 of 78 frauds at 15% precision, matching offline results; replaying a 5,011-purchase card history reproduced training features exactly.
- **Data quality over model performance:** Diagnosed concept drift (online fraud fell 66.4 → 0.0 per 10,000 after April 2017), removed the drifted feature, and recovered top-1,000 frauds from 17 to 150.

Dataset Quality Audits — Crop Yield & Used-Car Price Models (deployed)

- **Outlier & unit audit:** Caught unlabelled non-tonne yield units and impossible outliers; fixes cut test MAE from 107.63 to 2.04 t/ha across 246,091 government rows.
- **Error-driven data curation:** Traced the main error on ~15,400 car listings to one ₹3.95-crore listing and removed brands with under 10 listings.

EDUCATION

Bachelor of Engineering, Computer Science and Engineering | Chandigarh University, Mohali | 2026

CERTIFICATIONS

- Mastering Model Mechanics (AI Agent Evaluation) — Mercor (2026)
- Introduction to Artificial Intelligence (AI) — IBM, Coursera (2025) | Verify: coursera.org/verify/THQYI9YQTETZ
- Computer Vision Specialization — University of Colorado Boulder, Coursera (2026) | Verify: coursera.org/verify/specialization/TK90U2CA74AJ